

Course: Artificial Intelligence	ASSIGNMENT- 2	DUE DATE: 29/03/2026
Course Code: AI2002		Total Questions: 7 Total Marks: 100

Instructions:

1. Attempt **all questions by hand**.
2. **Scan your handwritten work** and submit as a **PDF** on GCS.
3. No plagiarism—copied work gets zero.

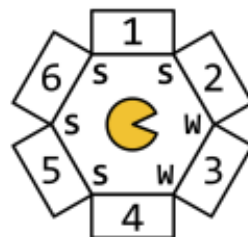
Question 1:

[15 Marks]

Q1. Pacman is trapped! He is surrounded by mysterious corridors, each of which leads to either a pit (P), a ghost (G), or an exit (E). In order to escape, he needs to figure out which corridors, if any, lead to an exit and freedom, rather than the certain doom of a pit or a ghost:

The one sign of what lies behind the corridors is the wind: a pit produces a strong breeze (S) and an exit produces a weak breeze (W), while a ghost doesn't produce any breeze at all. Unfortunately, Pacman cannot measure the strength of the breeze at a specific corridor. Instead, he can stand between two adjacent corridors and feel the max of the two breezes. For example, if he stands between a pit and an exit, he will sense a strong (S) breeze, while if he stands between an exit and a ghost, he will sense a weak (W) breeze. The measurements for all intersections are shown in the figure below.

Also, while the total number of exits might be zero, one, or more, Pacman knows that two neighboring squares will not both be exits.



Pacman models this problem using variables X_i for each corridor i and domains P, G, and E

(a) [3] State the binary and/or unary constraints for this CSP (either implicitly or explicitly).

Binary:

$X_1 = P$ or $X_2 = P$, $X_2 = E$ or $X_3 = E$,

$X_3 = E$ or $X_4 = E$, $X_4 = P$ or $X_5 = P$,

$X_5 = P$ or $X_6 = P$, $X_1 = P$ or $X_6 = P$,

$\forall i, j$ s.t. $\text{Adj}(i, j) \rightarrow \neg(X_i = E \text{ and } X_j = E)$

Unary:

$X_2 \neq P$, $X_3 \neq P$, $X_4 \neq P$

Note: This is just one of many solutions. The answers below will be based on this

formulation

- (b) [3] Suppose we assign X_1 to E. Perform forward checking after this assignment. Also, enforce unary constraints.

X_1			E
X_2			
X_3		G	E
X_4		G	E
X_5	P	G	E
X_6	P		

- (c) [3] Suppose forward checking returns the following set of possible assignments:

X_1			E
X_2			
X_3		G	E
X_4		G	E
X_5	P		
X_6	P		

According to MRV, which variable or variables could the solver assign first?

- (d) [3] Assume that Pacman knows that $X_6 = G$. List all the solutions of this CSP or none if no solutions exist.

X_1	P		
X_2		G	E
X_3		G	E
X_4		G	E
X_5	P		
X_6	P	G	E

According to MRV, which variable or variables could the solver assign first?

X1 or X5 (tie breaking)

- (e) [3] Assume that Pacman knows that $X_6 = G$. List all the solutions of this CSP or none if no solutions exist.

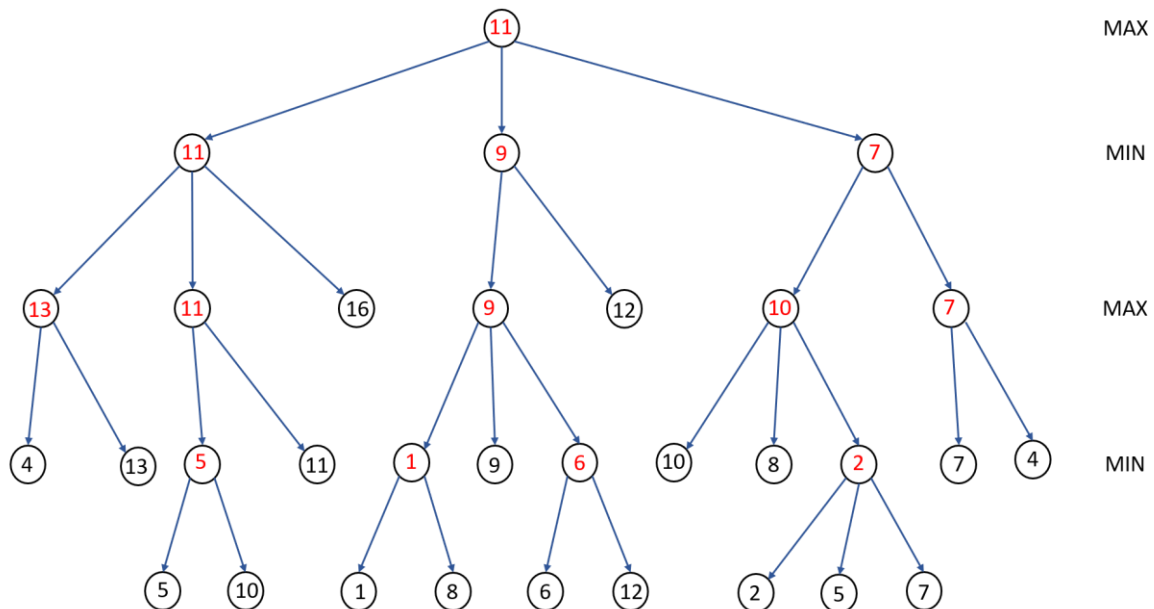
(P,E,G,E,P,G)

(P,G,E,G,P,G)

Question 2:

[15 Marks]

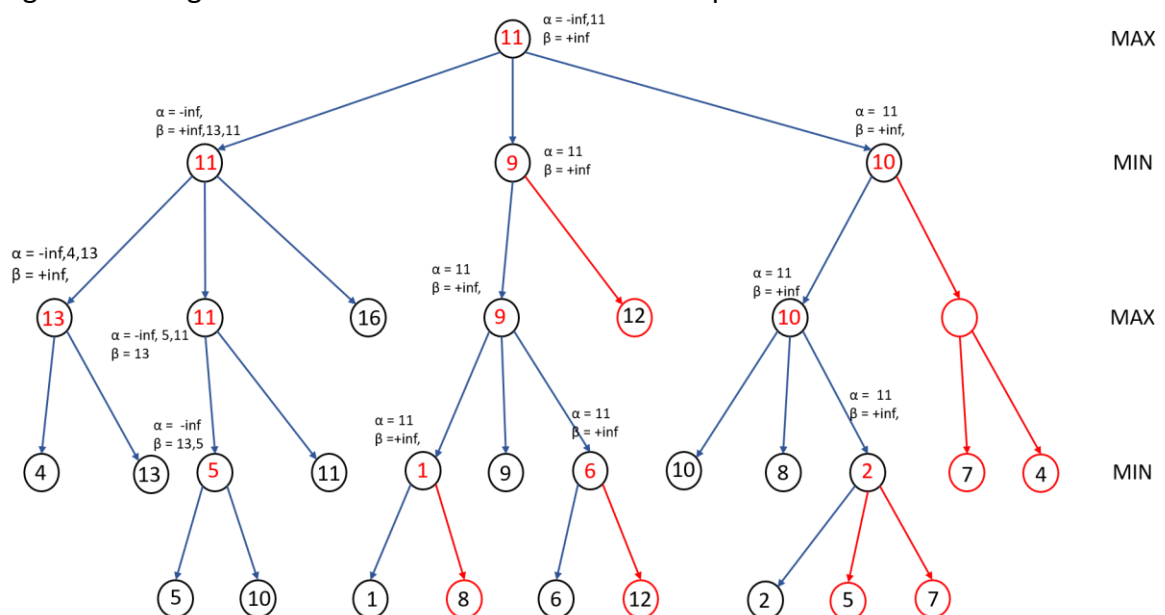
- (a) [4] Use the Minimax algorithm to compute the minimax value at each node for the game tree below.



[1] for having the correct value at the root node

[-0.25] for each incorrect value at the inner nodes (deduct a maximum of 3 points)

- (b) [9] Use the Alpha-Beta pruning algorithm to prune the game tree above assuming child nodes are visited from left to right. Show all final alpha and beta values computed at the root, each internal node visited, and at the top of pruned branches. Note: Follow the algorithm in Figure 5.7 in the textbook. Also show the pruned branches.



[2] for having the correct alpha and beta values at the root node

- [1] for the alpha value
- [1] for the beta value

[5] for having the correct alpha and beta values at the inner nodes right before the pruned nodes (5 of them in total)

- [-1] for each incorrectly marked pruned node
- [0.5] for each of the alpha values
- [0.5] for each of the beta values

[2] for having the correct alpha and beta values at the other inner nodes [deduct maximum of 2 points]

- [0.25] for each of the alpha values
- [0.25] for each of the beta values

(c) [2] For a general game tree (i.e., not limited to the above tree), are there any cases that the Alpha-Beta algorithm gives a different value at the root node than the Minimax algorithm? If yes, show an example; if no, just say no.

[2] No

Question 3:

[15 Marks]

Suppose three documents 1, 2 and 3 contain only five words “I”, “am”, “we”, “are” and “groot”. The following table summarizes the number of occurrences of each word in each document. Suppose one document is chosen at random (each document with equal probability), then one word in that document is chosen at random (each word with equal probability within that document).

Document / Word	I	am	we	are	groot
1	12	12	1	1	13
2	17	17	0	0	17
3	14	14	2	2	15

Note that the table can be used to compute conditional probabilities. For example, $P(\text{Word} = \text{groot} \mid \text{Document} = 1) = \frac{13}{12 + 12 + 1 + 1 + 13}$

- [2] What is the probability that the word is “groot”, i.e., $P(\text{Word} = \text{groot})$?
- [2] Suppose the randomly chosen word is “we”. What is the probability that it comes from document 1, i.e., $P(\text{Document} = 1 \mid \text{Word} = \text{we})$?
- [2] Suppose the randomly chosen word starts with an “a”. What is the probability that it comes from document 2, i.e., $P(\text{Document} = 2 \mid \text{Word} = \text{am or Word} = \text{are})$?

Now suppose document 1 is chosen with probability $P(\text{Document} = 1) = \frac{1}{6}$, document 2 is chosen with probability $P(\text{Document} = 2) = \frac{1}{3}$, and document 3 is chosen with probability $P(\text{Document} = 3) = \frac{1}{2}$. Then, one word in that document is chosen at random (each word with equal probability within that document).

- (d) [3] What is the probability that the word is “groot”, i.e., $P(\text{Word} = \text{groot})$?
- (e) [3] Suppose the randomly chosen word is “we”. What is the probability that it comes from document 1, i.e., $P(\text{Document} = 1 \mid \text{Word} = \text{we})$?
- (f) [3] Suppose the randomly chosen word starts with an “a”. What is the probability that it comes from document 2, i.e., $P(\text{Document} = 2 \mid \text{Word} = \text{am or Word} = \text{are})$?

Show your calculations and round all final answers to 4 decimal places.

Answers:

Document \ Word	I	Am	We	Are	Groot	Total
1	12	12	1	1	13	39
2	17	17	0	0	17	51
3	14	14	2	2	15	47

$$(a) \frac{1}{3} \left(\frac{13}{39} + \frac{17}{51} + \frac{15}{47} \right) \approx 0.3286$$

$$(b) \frac{\frac{1}{39}}{\frac{1}{39} + \frac{0}{51} + \frac{2}{47}} = 0.3760$$

$$(c) \frac{\frac{17}{51}}{\frac{13}{39} + \frac{17}{51} + \frac{16}{47}} \approx 0.3310$$

$$(d) \frac{1}{6} \frac{13}{39} + \frac{1}{3} \frac{17}{51} + \frac{1}{2} \frac{15}{47} \approx 0.3262$$

$$(e) \frac{\frac{1}{6} \frac{1}{39}}{\frac{1}{6} \frac{1}{39} + \frac{1}{3} \frac{0}{51} + \frac{1}{2} \frac{2}{47}} \approx 0.1673$$

$$(f) \frac{\frac{1}{3} \frac{17}{51}}{\frac{1}{6} \frac{13}{39} + \frac{1}{3} \frac{17}{51} + \frac{1}{2} \frac{16}{47}} \approx 0.3298$$

Question 4:

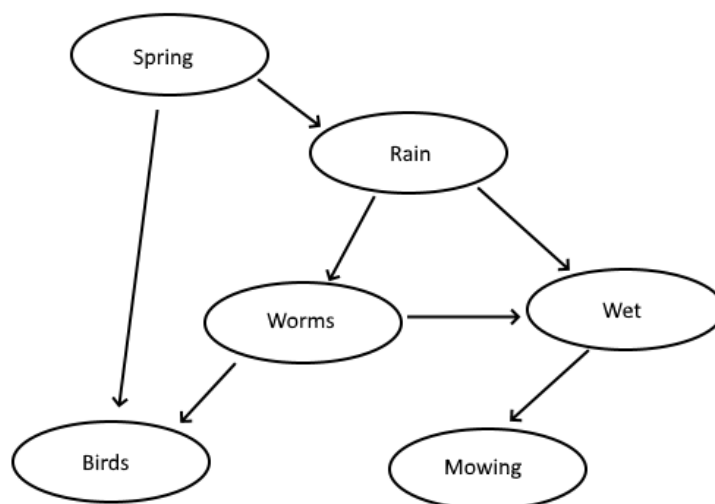
[20 Marks]

Use the Bayesian Network below containing 6 Boolean random variables to answer the following questions using inference by enumeration. For the problems, Spring = S, Rain = R, Worms = Wo, Wet = We, Mowing = M, and Birds = B. Give your answers to 4 decimal places. Show your work.

- [4] What is the probability that it rains? ($P(R)$)
- [4] What is the probability that the grass is wet? ($P(We)$)
- [4] Given that it is spring, how likely is there to be someone mowing? ($P(M \mid S)$)
- [4] Given that it is spring, how likely is there to be a bird on the lawn? ($P(B \mid S)$)
- [4] If there are birds on the lawn, but there are no worms out, how likely is it that it is spring? ($P(S \mid B, \neg Wo)$)

The CPTs are defined as follows:

- $P(S) = 0.25$
- $P(R \mid S) = 0.7, P(R \mid \neg S) = 0.3$
- $P(Wo \mid R) = 0.7, P(Wo \mid \neg R) = 0.2$
- $P(We \mid R, Wo) = 0.12, P(We \mid R, \neg Wo) = 0.25, P(We \mid \neg R, Wo) = 0.1, P(We \mid \neg R, \neg Wo) = 0.08$
- $P(M \mid We) = 0.02, P(M \mid \neg We) = 0.42$
- $P(B \mid S, Wo) = 0.8, P(B \mid S, \neg Wo) = 0.4, P(B \mid \neg S, Wo) = 0.4, P(B \mid \neg S, \neg Wo) = 0.4$



$$(a) P(R) = P(R|S) \cdot P(S) + P(R|\neg S) \cdot P(\neg S) = 0.7 \cdot 0.25 + 0.3 \cdot 0.75 = 0.4$$

$$(b) P(We) = P(We|R, Wo) \cdot P(Wo|R) \cdot P(R) + P(We|R, \neg Wo) \cdot (1 - P(Wo|R)) \cdot P(R) + P(We|\neg R, Wo) \cdot P(Wo|\neg R) \cdot (1 - P(R)) + P(We|\neg R, \neg Wo) \cdot (1 - P(Wo|\neg R)) \cdot (1 - P(R)) = 0.12 \cdot 0.7 \cdot 0.4 + 0.25 \cdot 0.3 \cdot 0.4 + 0.1 \cdot 0.2 \cdot 0.6 + 0.08 \cdot 0.8 \cdot 0.6 = 0.114$$

$$(c) P(M|S): \text{This value is useful first } \{P(We|S) = P(We|R, Wo) \cdot P(Wo|R) \cdot P(R|S) + P(We|R, \neg Wo) \cdot (1 - P(Wo|R)) \cdot P(R|S) + P(We|\neg R, Wo) \cdot P(Wo|\neg R) \cdot (1 - P(R|S)) + P(We|\neg R, \neg Wo) \cdot (1 - P(Wo|\neg R)) \cdot (1 - P(R|S)) = 0.12 \cdot 0.7 \cdot 0.7 + 0.25 \cdot 0.3 \cdot 0.7 + 0.1 \cdot 0.2 \cdot 0.3 + 0.08 \cdot 0.8 \cdot 0.3 = 0.1365\}; P(M|S) = \frac{P(M,S)}{P(S)} = \frac{P(M|We)P(We|S)P(S) + P(M|\neg We)(1 - P(We|S))P(S)}{P(S)} = P(M|We) \cdot P(We|S) + P(M|\neg We) \cdot (1 - P(We|S)) = 0.02 \cdot 0.1365 + 0.42 \cdot 0.8635 = 0.3654$$

$$(d) P(B|S) \text{ This value is useful first } \{P(Wo|S) = P(Wo|R) \cdot P(R|S) + P(Wo|\neg R) \cdot (1 - P(R|S)) = 0.7 \cdot 0.7 + 0.2 \cdot 0.3 = 0.55\}; P(B|S) = P(B|S, Wo) \cdot P(Wo|S) + P(B|S, \neg Wo) \cdot (1 - P(Wo|S)) = 0.8 \cdot 0.55 + 0.4 \cdot 0.45 = 0.62$$

$$(e) P(S|B, \neg Wo) \text{ This value is useful first } \{P(Wo|\neg S) = P(Wo|R) \cdot P(R|\neg S) + P(Wo|\neg R) \cdot (1 - P(R|\neg S)) = 0.7 \cdot 0.3 + 0.2 \cdot 0.7 = 0.35\}; P(S|B, \neg Wo) = \frac{P(B,S,\neg Wo)}{P(B,\neg Wo)} = \frac{P(S,\neg Wo)[(1 - P(R))P(S)P(S) + (1 - P(\neg R))(1 - P(S))P(S)]}{P(S,\neg Wo)(1 - P(S))P(S) + P(\neg S,\neg Wo)(1 - P(\neg S))P(\neg S)} = \frac{0.4[(1 - 0.7) \cdot 0.7 \cdot 0.25 + (1 - 0.2)(1 - 0.7) \cdot 0.25]}{0.4(1 - 0.55)0.25 + 0.4(1 - 0.35) \cdot 0.75} = \frac{0.045}{0.24} = 0.1875$$

Question 5:

[15 Marks]

About 2/3 of your email is spam so you downloaded an open-source spam filter based on word occurrences that uses the Naive Bayes classifier. Assume you collected the following regular and spam mails to train the classifier, and only three words are informative for this classification, i.e., each email is represented as a 3-dimensional binary vector whose components indicate whether the respective word is contained in the email.

study	free	money	Category
1	0	0	Regular
0	0	1	Regular
1	0	0	Regular
1	1	0	Regular
0	1	0	Spam
0	1	0	Spam
0	1	0	Spam
0	1	0	Spam
0	1	1	Spam
0	1	1	Spam
0	1	1	Spam
0	1	1	Spam

- (a) [5] You find that the spam filter uses a prior $p(\text{spam}) = 0.1$. Explain (in one sentence) why this might be sensible and give the following model parameters when estimated as maximum-likelihood with add-one smoothing (i.e., using pseudo counts of one).

It is worse for regular emails to be classified as spam than it is for spam email to be classified as regular email.

$$P(\text{study}|\text{spam}) = 1/10$$

$$P(\text{study}|\text{regular}) = 2/3$$

$$P(\text{free}|\text{spam}) = 9/10$$

$$P(\text{free}|\text{regular}) = 1/3$$

$$P(\text{money}|\text{spam}) = 1/2$$

$$P(\text{money}|\text{regular}) = 1/3$$

- (b) [5] Based on the prior and conditional probabilities above, give the model probability $P(\text{spam}|s)$ that the sentence $s = \text{"money for psychology study"}$ is spam

$$p(\text{spam}) P(\text{study}|\text{spam}) (1 - P(\text{free}|\text{spam})) (P(\text{money}|\text{spam})) =$$

$$p(\text{spam}) * 1/200 (1 - p(\text{spam})) P(\text{study}|\text{regular}) (1 - P(\text{free}|\text{regular})) (P(\text{money}|\text{regular}))$$

$$= (1 - p(\text{spam})) * 4/27 P(\text{spam}|s) = p(\text{spam})/200 / [p(\text{spam})/200 + (1 - p(\text{spam})) * 4/27] =$$

$$p(\text{spam})/200 / [p(\text{spam})/200 + 4/27 - p(\text{spam}) * 4/27] = p(\text{spam})/200 / [4/27 -$$

$$p(\text{spam}) * 773/5400] = .003736$$

- (c) [5] What should be the value of the prior $p(\text{spam})$ if we would like the above sentence to have the same probability as being spam as not spam, i.e., it would be classified as spam with probability 0.5?

$$0.5 = p(\text{spam})/200 / [4/27 - p(\text{spam}) * 773/5400] \text{ iff } 4/27 - p(\text{spam}) * 773/5400 = p(\text{spam})/100 \text{ iff } p(\text{spam}) = .9674$$

Question 6:

[10 Marks]

Ecology explores relationships between organisms and other living (biotic) things or non-living (abiotic) components in their environment. Certain factors may impact the abundance, distribution, or physiology of organisms, including abiotic factors such as temperature, moisture, or sunlight, and biotic factors such as the presence of predators or competitors. Linear regressions can be used to quantify these relationships and predict organism responses to various levels of the abiotic or biotic factor. Ecologists tested the grazing pressure of green crabs on clams. They constructed $1m^2$ cages and planted 300 clams in each cage. Two days later, they counted the number of remaining clams and recorded the data in the data below.

Number of Crabs	Number of Clams Remaining
2	137
4	70
2	184
5	0
4	35
0	297
3	122
5	1
1	253
3	150

First, perform a linear regression using the step-by-step instructions for calculating m (slope) and b (intercept) of the regression line. **What is the full equation for the regression line that you calculated?**

Step 1. Decide which variable is the independent variable and which is the dependent variable: In this example, the number of crabs is the independent variable (x), and the number of clams remaining is the dependent variable (y). This is because the researchers chose the number of crabs to place in each plot, and the resulting number of clams *depends* on how many crabs are present.

Step 2. Calculate x^2 for every value of x , and y^2 for every value of y . For each pair of values, calculate $x \times y$.

You should end up with ten values (the same number of values of x) for x^2 , ten values for y^2 , and ten values for $x \times y$.

x^2	y^2	$x \times y$
4	18,769	274
16	4,900	280
4	33,856	368
25	0	0
16	1,225	140
0	88,209	0

9	14,884	366
25	1	5
1	64,009	253
9	22,500	450

Step 3. Calculate $\sum x$, $\sum y$, $\sum(x)^2$, $\sum(y)^2$, and $\sum(x \times y)$.

$$\sum x = 2 + 4 + 2 + 5 + 4 + 0 + 3 + 5 + 1 + 3 = 29$$

$$\sum y = 137 + 70 + 184 + 0 + 35 + 297 + 122 + 1 + 253 + 150 = 1249$$

$$\sum(x^2) = 4 + 16 + 4 + 25 + 16 + 0 + 9 + 25 + 1 + 9 = 109$$

$$\sum(y^2) = 18769 + 4900 + 33856 + 0 + 1225 + 88209 + 14884 + 1 + 64009 + 22500 = 248353$$

$$\sum(x \times y) = 274 + 280 + 368 + 0 + 140 + 0 + 366 + 5 + 253 + 450 = 2136$$

$\sum x$	$\sum y$	$\sum(x^2)$	$\sum(y^2)$	$\sum(x \times y)$
29	1249	109	248,353	2136

Step 4. Familiarize yourself with the equation of a line: $y = mx + b$.

Step 5. Calculate m (slope):

$$\frac{n(\sum(x \times y)) - (\sum x)(\sum y)}{n(\sum(x^2)) - (\sum x)^2}$$

To simplify this calculation, let's start with the **numerator**: $(n(\sum(x \times y)) - (\sum x)(\sum y))$

Remember n ! In this example, $n = 10$ because we have 10 values of x and 10 values of y .

$$\text{numerator} = (10(2136) - (29)(1249))$$

$$\text{numerator} = (21360 - 36221)$$

$$\text{numerator} = -14861$$

Now we can calculate the bottom half of the equation (this is called the **denominator**): $(n(\sum(x^2)) - (\sum x)^2)$

$$\text{denominator} = (10(109) - (29)^2)$$

$$\text{denominator} = (1090 - 841)$$

$$\text{denominator} = 249$$

To finish the calculation, divide the numerator by the denominator:

$$m = -\frac{14861}{249} = -59.683$$

Here, we have rounded the actual answer from a long decimal string (-59.68273092...) to **-59.683**.

Step 6. Calculate b (intercept):
$$\frac{\Sigma y - m(\Sigma x)}{n}$$

Again we can start on the **numerator** (top half of the equation): $(\Sigma y - m(\Sigma x))$

$$\text{numerator} = (1249 - (-59.683)(29))$$

$$\text{numerator} = (1249 - (-1730.807))$$

$$\text{numerator} = 2979.807$$

Next, we would calculate the **denominator** (bottom half of the equation), but this is just n !

$$\text{denominator} = 10$$

To finish the calculation, divide the numerator by the denominator:

$$b = \frac{2979.807}{10} = 297.98$$

Again, we have rounded the actual answer from a longer decimal (297.9807) to **297.98**.

Step 7. Replace ' b ' and ' m ' in the line equation: $y = mx + b$.

From the previous steps, we know:

$$m = -59.683$$

$$b = 297.98$$

So the final equation of the linear regression is: $y = -59.683x + 297.98$

Question 7:

[10 Marks]

A hospital wants to predict the likelihood of patients developing complications after a surgical procedure based on various factors such as age, pre-existing health conditions, and type of surgery. They have collected data on 500 patients who underwent the same surgical procedure in the past year. For each patient, they recorded their age, presence of diabetes, presence of hypertension, whether they smoke, and the type of surgery performed (coded as 1 for minor surgery and 2 for major surgery), as well as whether they experienced complications post-surgery (coded as 1 for yes and 0 for no).

Age	Diabetes	Hypertension	Smoker	Surgery Type	Complications
45	0	1	1	1	0
60	1	1	0	2	1
55	1	0	1	1	1
70	0	1	0	2	1
50	1	1	1	1	0

use the model to predict the probability of complications for a new patient who is 60 years old, has diabetes and hypertension, is a smoker, and is undergoing major surgery.

Step 1: Logistic Regression Model

The logistic regression model is:

$$P(Y = 1) = \frac{1}{1 + e^{-(b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 + b_5x_5)}}$$

where

- x_1 = Age
- x_2 = Diabetes
- x_3 = Hypertension
- x_4 = Smoker
- x_5 = Surgery Type

$$z = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 + b_5x_5$$

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

Step 2: Example Coefficients

Assume the trained logistic regression model produced these coefficients (illustrative example):

Variable	Coefficient
Intercept (b_0)	-8
Age (b_1)	0.05
Diabetes (b_2)	1.5
Hypertension (b_3)	1.2

Smoker (b4)	1.0
Surgery Type (b5)	2.0

Step 3: New Patient Information

Feature	Value
Age	60
Diabetes	1
Hypertension	1
Smoker	1
Surgery Type	2

Step 4: Compute Linear Combination (z)

$$z = -8 + (0.05 \times 60) + (1.5 \times 1) + (1.2 \times 1) + (1.0 \times 1) + (2 \times 2)$$
$$z = -8 + 3 + 1.5 + 1.2 + 1 + 4, z = 2.7$$

Step 5: Convert to Probability

$$P(Y = 1) = 1 / (1 + e^{-2.7})$$

$$e^{-2.7} = 0.067$$

$$P(Y = 1) = 1 / (1 + 0.067)$$

$$P(Y = 1) = 0.937$$